INCH-POUND

MIL-C-29627(NU)
31 January 90

MILITARY SPECIFICATION

COVERALLS, CATAPULT CREWMEN'S (FLAME RESISTANT)

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a utility coverall for catapult crewmen fabricated from a flame resistant, plain weave, aramid cloth.

1.2 Classification. The coveralls shall be of one type only and in the following sizes (see 6.2):

Schedule of Sizes

Small (36-38)	Large (44-46)
Medium (40-42)	X-Large (48-50)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

KK-L-2004	- Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
DDD-L-20	- Label; For Clothing, Equipage and Tentage (General Use)

MILITARY

MIL-C-3735	- Cuffs, Knit, Wrist and Ankle, and Cloth, Knitted
MIL-F-10884	- Fasteners, Snap
MIL-P-15011	- Pallet, Material Handling, Wood Post Contruction, 4 Way Entry
MIL-B-17757	- Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-F-21840	- Fastener Tapes, Hook and Loop, Synthetic
MIL-T-44100	- Thread, Aramid, Spun, Intermediate Modulus
MIL-T-83193	- Thread, Aramid, Spun Staple
MIL-C-83429	- Cloth, Plain and Basket Weave, Aramid

STANDARDS

FEDERAL

FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection for Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-1668	- Provisions for Evaluating Quality of Cloth Coveralls
MIL-STD-2073-1	- DOD Materials Procedures for Development and Application of Packaging Requirements
MIL-STD-2073-2	- Packaging Requirement Codes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

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3.3 Material.

3.3.1 Basic material. The basic fabric shall be a plain weave, aramid cloth conforming to type II, class 1 of MIL-C-83429. The color shall be Kelley Green shade 3423.

3.3.2 Leather. The leather for knee patches shall be a flesh split cattlehide, natural color of tannage, of not less than 2-3/4 nor more than 3-1/2 ounce thickness conforming to type IV, Class 2 of KK-L-2004 after silicone treatment specified in 3.3.2.1, except for the following:

a. Shrinkage temperature shall be not less than 105°C.
b. Tests for percent ash and chloroform-soluble material are not required.

3.3.2.1 Silicone treatment. The leather shall be treated with an approved durable silicone compound (see 6.5) to lower its water absorption. The treated leather shall not absorb more than 30 percent water when tested as specified in 4.4.1.

3.3.3 Knit wristlets. The aramid knit wristlets shall conform to type IV (Aramid), class 2 of MIL-C-3735. The color shall be either olive green or sage green.

3.3.4 Fastener tapes. The nylon fastener tape shall be flame-retardant coated conforming to type II, class 1 of MIL-F-21840. The color shall be Black 3230. The finished width shall be 1 inch. All cut lengths shall be 2 inches long except for two 4-inch long loop fastener tapes for the back leg.

3.3.5 Thread. The thread shall be aramid thread, size T-45, 3 ply, conforming to MIL-T-83193. As an alternate, a para-aramid, spun staple thread, size T-35, 2 ply, conforming to MIL-T-44100 may be used. The color of the thread to be used for the coverall shall approximate the basic fabric. The color of the thread to be used for applying the reflective tape (see 3.3.8) shall approximate the reflective tape. The thread shall meet commercially acceptable colorfastness requirements for aramid thread.

3.3.6 Fasteners, snap. The snap fasteners for the waistline closure shall be brass, conforming to style 2A, finish 2, male and female complete, consisting of stud with size 1 or 2 eyelet and socket with size 1 or 2 eyelet (instead of button) conforming to MIL-F-10884.

3.3.7 Webbing, elastic. The elastic webbing for the back waistband channel shall be 100% polyester webbing, 1 inch wide, Elastic Corporation of America style #114, or equal (see 6.6 and 6.9).

3.3.8 Reflective tape. The 1/2 inch wide reflective tape for shoulder, sleeve, and wrist shall be 3M Company style #8930 gray, or equal (see 6.7 and 6.9).

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3.3.9 Neoprene foam. The neoprene foam for the knee patch padding shall be a closed cell, chemically expanded, neoprene rubber, pigmented black, 3/32 (+ 1/64) inch thick, skin on one side and 901 black nylon on the other side, conforming to Rubatex Corp. style #SIS/901 Black NIS, or equal (see 6.8 and 6.9).

3.3.10 Labels.

3.3.10.1 Combination identification/instruction label. Each coverall shall have a combination identification/instruction label conforming to Type VI, class 15 of DDD-L-20. The printing shall show fastness to laundering. The label shall bear the following inscription:

COVERALLS, CATAPULT, CREWMEN'S (FLAME RESISTANT)
BASIC MATERIAL: 100% ARAMID
CONTRACT NO. DLA-000-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (If other than contractor)

LAUNDRY INSTRUCTIONS

MACHINE WASH, HOT WATER
DO NOT USE CHLORINE BLEACH
TUMBLE DRY, MEDIUM HEAT
REMOVE PROMPTLY

OR

USE NAVEDTRA 414-01-45-81 FORMULA II

DO NOT STARCH

3.3.10.2 Size label. Each coverall shall have a size label conforming to Type VI, class 2 of DDD-L-20. The printing shall show fastness to laundering. The label shall bear the following inscription:

SIZE: MEDIUM (40-42) (EXAMPLE)
STOCK NUMBER: 8415-00-000-0000 (EXAMPLE)

3.4 Design. The coverall shall have hook and loop fastener tapes for the front fly closure, a collar stand neckline, and elastic webbing in the channel back waistband. The sleeve shall have a self-fabric elbow patch and a ribbed knit cuff. The leg shall have a leather knee patch and an adjustable ankle closure of hook and loop fastener tape attached to a tab. The shoulder, arm, and lower sleeve areas shall have 1/2 inch wide reflective tape, as shown in Figure 1.

3.4.1 Figure. The figure is furnished for information only. If any inconsistency between the specification and figure occur, the specification shall govern.

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3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are permitted on the working patterns. Also, minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the appearance, serviceablity, or dimensional requirements cited in the specification. The patterns provide a 1/2 inch seam allowance for all joining seams unless otherwise indicated in Table I.

3.5.1 Pattern parts. The component parts of the coverall shall be cut from the material as specified below:

Material	Nomenclature of pattern part	Cut parts
Basic material	Upper front	2
	Upper back	1
	Sleeve	2
	Elbow patch	2
	Collar	2
	Waistband	1
	Lower front	2
	Lower back	2
	Leg tab	2
Leather	Knee patch	2
Neoprene foam	Knee patch	2
Marker	Front hook and loop location	-

3.6 Construction.

3.6.1 Stitches, seams, and stitchings. Stitch, seam and stitching types specified in Table I shall conform to FED-STD-751. Wherever two or more methods for seam or stitch types are given for the same part of the operation, any one of them may be used. Seam allowances shall be maintained with seam sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Where stitch type 401 is used, the looper or underthread shall be on the inside of the coveralls. Ends of all seams produced by stitch type 401 shall be caught in other seams or stitching. The bight of overedge stitching shall be 3/16 inch. The guide and knife shall be set to trim only the ravelled ends of the fabric. Double needle stitching shall be 1/4 (± 1/16) inch gauge, with the outer row of stitching 1/16 to 1/8 inch from the finished edge, unless otherwise specified. Hook and loop fastener tape stitching shall be 1/16 to 1/8 inch from edge of tape. Bartacks shall be 3/8 to 1/2 inch long and the width shall be 1/8 (± 1/32) inch wide.

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3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks (401 stitch type) shall be repaired by using 301 stitch type. The seat seam shall be completely resewn. Where stitch type 401 is permitted as an option, it shall not be used unless the ends are caught in other seams or stitchings.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.7 Manufacturing operations requirements. The items shall be manufactured in accordance with operation requirements as specified in Table I. The contractor is not required to follow the exact sequence of operations provided the finished items are identical to those produced by following the sequence of operations as listed in Table I. Minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the appearance, serviceability, or dimensional requirements cited in the specification.

3.7.1 Size marking. The component parts of the coveralls shall be marked, ticketed or bundled to insure a uniform size throughout the garment. Any method may be used except:

(1) Corrosive metal fastening devices.
(2) Sew-on shade tickets.
(3) Adhesive type tickets which discolor or adhere to the material upon removal of the tickets.

NOTE: The use of ink pad numbering machines, rubber stamps, or pencil is allowed, provided the numbering does not show on the outside and wherever possible, is covered by the seam allowance.

3.7.2 Abbreviations in table of operations. The abbeviations used in Table I are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
smlr	-	similar
Btnhl	-	Buttonhole
incl	-	including
dbl	-	double

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TABLE I - CONSTRUCTION OF COVERALLS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
1.	Cutting. a. The coveralls shall be cut in strict accordance with patterns furnished, which show size, shape and directional lines for cutting, and notches for proper assembly of all parts. The use of holding devices such as staples, etc., on the paper pattern parts which puncture the base fabric is prohibited.				See 3.3.5
	b. Directional lines on the fronts and backs of the coverall trouser parts may vary from the warp direction by not more than 1-1/2 inches. Measurements shall be taken from the top and bottom of the directional line on pattern to the selvage edge (on the even side of lay) of the fabric and the difference between the two measurements shall not exceed 1-1/2 inches.				
	c. Cut all component parts from the same piece of material except the leg tabs, which may be cut from ends and when so cut shall approximate the main assembly.				
	d. The waistband may be stripped and placed on rolls.				
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1668 shall be removed from production and replaced with non-defective and properly matched components.				
3.	Marking. a. All parts shall be marked, ticketed, or bundled to assure a uniform shade, size and proper assembly throughout the coverall. Stripped waistband and parts cut from ends need not be marked or ticketed, however, they shall approximate the main assembly.				

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TABLE I - CONSTRUCTION OF COVERALLS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
4.	Make leg tabs.				
	a. Fold leg tabs in half lengthwise, face to face and stitch 1/4 inch seam allowance the length of the tab. As an alternate, stitch 1/4 inch seam allowance the length of the tab and on one end. Turn to finished position.	301	SSe-2(a)	8-12	
	b. Edgestitch 1/8 to 1/16 inch on top and bottom folded edges.	301	SSe-2(b)	8-12	
	c. As an alternate to operations 4a and 4b, fold leg tab in half lengthwise, back to back. Turn a 1/4 inch seam allowance in on the length of the tab and edgestitch 1/8 to 1/16 inch on both top and bottom folded edges.	301	SSc-1	8-12	
	d. The finished width of the tab shall be 1-1/4 (± 1/8) inch.				
	e. Fold one raw end of leg tab under 1 (+1/16) inch and stitch all four sides of 2 (± 1/4) inch long hook fastener tape to back side of tab according to marks on pattern.	301	LSbj-1 and EFa-1	8-12	
5.	Make collar.				
	a. Place collar pieces face to face and stitch 1/4 inch from top and side raw edges. Turn to finished position.	301	SSa-1	8-10	
6.	Attach elbow patch and reflective tape.				
	a. Position elbow patch on sleeve according to marks on pattern. Turn under 1/2 inch on top, bottom, and inside raw edges of elbow patch, leaving inseam unfinished. Double stitch turned edges to sleeve.	301 or 401	LSd-2	8-12	
	b. Position and double stitch 1/4 inch gauge the 1/2 inch reflective tape to lower sleeve according to marks on pattern.	301	SSaa-2	8-12	

NOTE: Reflective tape shall be positioned evenly and the lines on the tape shall meet on the finished seams.

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TABLE I - CONSTRUCTION OF COVERALLS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
7.	Attach knee patch.				
	Place the neoprene knee patch padding behind the leather knee patch with the skin side toward the wearer. Position knee patch on the outer shell in accordance with pattern marks and double stitch all sides.	301 or 401	LSbj-2	8-12	
8.	Join outseams on lower front and back.				
	a. Stitch lower front and back at outseam using a double lapped and double stitched seam, front over back.	301 or 401	LSc-2	8-12	
9.	Join seat seam.				
	a. Join the seat seam using a double lapped and double stitched seam. The right back shall be over the left back.	301 or 401	LSc-2	8-12	
10.	Join shoulder seams.				
	a. Join shoulder seams of upper fronts and back, face to face. Stitch 1/2 inch from raw edge.	301 or 401	LSq-2(a)	8-12	
	b. Overedge the raw edges together.	503 or 504 or 602	SSa-1	8-12	
	c. Operations 10a and 10b may be performed in one operation by simultaneously stitching and overedging 3/8 inch from edge.	515 or 516 or 519	SSa-2	8-12	
	d. Turn seam allowance to back of coveralls and topstitch 1/4 inch from fold, catching seam allowance.	301	LSq-2(b)	8-12	
11.	Set sleeve.				
	a. Set sleeve to armhole, face to face. Stitch 1/2 inch from raw edge.	301 or 401	LSq-2(a)	8-12	
	b. Overedge the raw edges together.	503 or 504 or 602	SSa-1	8-12	

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NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
	c. Operations 11a and 11b may be performed in one operation by simultaneously stitching and overedging 3/8 inch from edge.	515 or 516 or 519	SSa-2	8-12	
	d. Turn seam allowance into body of coveralls and topstitch 1/4 inch from fold, catching seam allowance.	301	LSq-2(b)	8-12	
12.	Apply reflective tape. Center and double stitch 1/4 inch gauge the 1/2 inch reflective tape from neck edge along shoulder seam and down sleeve center to bottom of sleeve.	301 or 401	SSaa-2	8-12	
13.	Join underarm and side seams. a. Join the underarm and side seams in one continuous operation using a double lapped and double stitched seam with the fronts over the backs, catching the elbow patch in the stitching.	301 or 401	LSc-2	8-12	
NOTE:	The seam at the armhole junction shall not be staggered more than 3/8 inch (measured from center of seam to center of seam).				
14.	Join knit wristlet to sleeve. a. Fold knit wristlet in half with raw edges even. Align raw edges of wristlet and sleeve, face to face, and stitch.	301 or 401	LSq-2(a)	8-12	
	b. Overedge the raw edges together.	503 or 504 or 602	SSa-1	8-12	
	c. Operations 14a and 14b may be performed in one operation by simultaneously stitching and overedging 3/8 inch from edge.	515 or 516 or 519	SSa-2	8-12	

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NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
15.	Attach upper body to lower body at waist.				
	a. Position upper body and lower body, back to back, at waist and stitch 1/2 inch from the raw edge. The seam allowance shall finish on the outside and shall be turned toward the upper body.	301 or 401	SSa-1	8-12	
	b. Overedge the raw edges together.	503 or 504 or 602	SSa-1	8-12	
	c. Operations 15a and 15b may be performed in one operation by simultaneously stitching and overedging 3/8 inch from the edge.	515 or 516 or 519	SSa-2	8-12	
16.	Attach webbing to back waist.				
	a. Position 1 inch wide polyester elastic webbing on outside of coverall at back side seams above the waist joining seam (see operation 15a), according to the following lengths:				
	Size — Length in Inches: Small 14; Medium 16; Large 18; X-Large 20				
	b. Stitch two 3/8 inch vertical bartacks through each end of webbing and side seams.	Brtck		28 per brtck	
17.	Attach waistband.				
	a. Position waistband so that when finished, it shall cover the waist joining seam and elastic of coveralls (see operation 16a).				
	b. Stitch lower edge of waistband to trouser portion, face to face, 1/4 inch from raw edge.	301 or 401	LSq-2(a)	8-12	

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NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
	c. Turn waistband to finished position, turning raw edge under 1/4 inch and edge stitch top and bottom folded edges being careful not to catch elastic.	301 or 401	LSq-2(b) and LSb-1	8-12	
	OR				
	d. As an alternate to operation 17b and 17c, the waistband may be attached by means of a double needle machine. The gauge shall be 1-1/2 (+ 1/8) inches between needles. The elastic shall not be caught in the stitching.	301 or 401	LSc-2	8-12	
	e. Waistband shall finish 1-3/4 (+ 1/16) inches wide.				
18.	Stitch crotch seam.				
	a. Overedge stitch each raw edge of the crotch seam allowance.	503 or 504 or 602	SSa-1	8-12	
	b. Stitch crotch seam 1/2 inch from the raw edge.	301 or 401	LSq-3(a)	8-12	
19.	Finish front facing and attach snap fastener closure.				
	a. Right front facing:				
	(1) - Overedge stitch grown on facing including lower edge.	503 or 504 or 602	EFd-1	8-12	
	(2) - Fold grown on facing, face to face, according to marks on pattern, and stitch at neckline only. Turn facing to finished position, forcing out point.	301	SSa-1	8-12	
	(3) - Stitch 1/8 to 3/16 inch from inside finished edge of facing.	301	EFa-1	8-12	

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NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
	(4) - Securely attach snap fastener (stud and post) 3/4 inch from folded edge of facing, centered on the waistband, through all plies of facing and waistband.				
NOTE:	The above measurement is taken from the center of the snap fastener.				
	b. Left front facing:				
	(1) - Overedge stitch lower edge of left front facing.	503 or 504 or 602	EFd-1	8-12	
	(2) - Turn left front facing back to back on first fold line of pattern marks. Center on waistband, 7/8 (+ 1/8) inch from folded edge of left facing, and secure snap fastener (cap and socket) through the two plies of facing and waistband.				
	(3) - Turn under left front facing on second fold line of pattern marks. Stitch the length of the left front, 1/16 to 1/8 inch from inside folded edge. Snap fastener cap and socket on left facing shall be aligned with snap fastener stud and post on right facing.	301	EFaa-1	8-12	
	c. Turn front crotch seam to the left and lap left facing over right facing. Topstitch the crotch seam catching the seam allowance and continue across the bottom of the facings, ending at previous left front facing stitchline (see operation 19b (3)).	301	LSq-2(b)	8-12	
	d. Superimpose a second row of topstitching on crotch seam, catching seam allowance (see operation 19c) and continue stitching 3/16 to 1/4 inch above previous stitching and ending as before.	301	LSq-2(b)	8-12	
NOTE:	Lower stitching at bottom of facings shall be no more than 3/8 to 1/2 inch from overedge edge.				

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NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
20.	Attach collar and size label.				
	a. Position face of collar to inside of neckline. Stitch raw edge of inner collar to neckline, inserting size label approximately centered under center back of collar.	301	LScf-2(a)	8-12	
	b. Turn collar to finished position, turning in lower edge of outer collar 3/8 inch. Edge stitch, through all plies of material, 1/16 inch from folded edge.	301	LScf-2(b)	8-12	
	c. Topstitch collar 3/16 to 1/4 inch from the outer edge.	301	SSe-2(b)	8-12	
21.	Attach hook and loop fastener tape to front facing.				
	a. Stitch all four sides of 2 inch long loop fastener tape to inside left front facing according to marker.	301	SSab-1	8-12	
	b. Stitch all four sides of 2 inch long hook fastener tape to outside right front facing according to marker, lining up hook fastener tape to correspond to loop fastener tape on left front facing.	301	SSab-1	8-12	
22.	Join inseam.				
	a. Join inseam in one continuous operation with a double lapped and double stitched seam with fronts over backs. The seam at the crotch junction shall not be staggered more than 3/8 inch (measured from center of seam to center of seam).	301 or 401	LSc-2	8-12	
23.	Attach leg tabs and loop fastener tape.				
	a. Stitch all four sides of a 2 inch long loop fastener tape to leg front and a 4 inch long loop fastener tape to leg back according to marks on the pattern.	301	LSbj-1	8-12	
	b. Fold unfinished end of leg tab under 1/2 inch and boxstitch tab to leg as marked on pattern having fastener tape end toward inseam. If alternate leg tab construction was used (see operation 4a), superimpose leg tab on leg as marked on pattern. The finished leg tab shall measure 4-3/4 (± 1/4) inches.	301	LSd-1	8-12	

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TABLE I - CONSTRUCTION OF COVERALLS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
24.	Hem leg bottoms.				
	a. Turn raw edge under 1/2 (± 1/8) inch. Fold back over body of material and stitch 1/16 inch from turned raw edge.	301	EFb-1	8-12	
	b. Hem shall finish 1 (± 1/8) inch in width.				
25.	Combination label.				
	a. Position top of combination label 2 to 3 inches down from neckline seam, centered (1/2 inch tolerance) on inside of back piece. Stitch all four sides.	301	LSbj-1	8-12	
26.	Bartacks.				
	All bartacks shall be 3/8 to 1/2 inch long by 1/8 (± 1/32) inch wide and located as follows:	Brtck		28 per brtck	
	- One at intersection of right collar and inner edge of right facing, placed at an angle to catch both collar and facing.				
	- Two on leg tab, vertically superimposed on the inside of the boxstitch.				
27.	Cleaning.				
	a. Trim all thread ends and thread floats on the outside of the coverall to 1/4 inch or less and on the inside to 1/2 inch or less.				
	b. Remove all loose threads from inside and outside of the coveralls.				
	NOTE: Thread trimming and removal may be performed after each operation or upon completion of all sewing operations. Caution shall be used so not to harm the material when trimming.				
	c. Remove all spots, stains, and shade tickets.				

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3.8 Sizes and measurements. The sizes and measurements of the finished coveralls shall be as shown in Table II. All measurements and tolerances are expressed in inches.

Table II - Sizes and measurements

Size	Chest A	Waist B	Hip C	Leg Inseam D	Sleeve Inseam E
Small (36-38)	46	34	44	29	17
Medium (40-42)	49	37	48	30	18
Large (44-46)	52	40	52	31	19
X-Large (48-50)	55	43	56	32	20
Tolerance	±1	±1	±1	±1/2	±1/2

NOTE: A, B, C, D, and E measurements refer to Figure 1. The coverall shall be laid flat with all front fasteners completely closed.

A. Chest - Measure across front at base of armholes from folded edge to folded edge. Double the value obtained.

B. Waist - Measure across the back waist taking care not to extend elastic. Double the value obtained.

C. Hip - Measure eight (8) inches down from lower edge of waistband on center back seam taking care that measurement is parallel to waistband. Double the value obtained.

D. Leg Inseam - Measure from crotch seam along inseam to bottom of hemmed leg.

E. Sleeve Inseam - Measure from base of armhole along inseam to wristlet seam. Do not include wristlet.

3.9 Workmanship. The finished coverall shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable "point values" or defect limit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2 shall be inspected for compliance with design, construction, workmanship, appearance, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-1668, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be acceptable for the identification of the silicone treatment of 3.3.2.1. In addition, other components shall be tested for the characteristics specified below and in accordance with the referenced test method of FED-STD-311, unless otherwise specified. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be in accordance with the following:

Lot size	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified.

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The basis for lot size and sample units for testing shall be as follows:

Component	Lot size expressed as	Sample unit for testing
Leather	Square feet	1/2 square foot

Table III - Testing of components

Component	Characteristic	Requirement paragraph	Test method
Leather	Water absorption	3.3.2.1	8111 of FED-STD-311

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with MIL-STD-1668 and the following defects:

DEFECT	POINT VALUE
LEATHER	
a. Not type specified	2
b. Any brittle or bony area, weak or thin spot	2
c. Leather improperly cut, excessive stretch	1
d. Wrinkles or loose, spongy or boardy leather	1
e. Any hole (except needlehole or pinhole), cut, or tear	2
REFLECTIVE TAPE	
a. One section missing	2
b. More than one section missing	3
c. Edge of tape not parallel with bottom edge of coverall cuff by more than 1/2 inch	1
d. Edge of tape lifting off fabric	3
ELBOW PATCH(S) AND KNEE PAD(S)	
a. Misplaced by more than 3/4 inch from pattern marks	1
b. Puckered, twisted, pleated, or poorly shaped, affecting appearance	1

4.4.3 Examination of packaging requirements. Examination of packaging shall be in accordance with MIL-STD-1668.

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4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each coverall shall be folded as follows:

1. Secure closures and snap fastener.
2. Lay garment flat, face down.
3. Fold garment in half, lengthwise, with the crotch inside the fold.
4. Fold sleeves over body with the sleeve at the center of the folded garment.
5. Bring bottom of trousers over knee patch.
6. Using the top of the knee patch as a guide, fold twice across the length of the garment.
7. Fold top of garment over the previous folds.

The folded garment shall measure approximately 11-1/2 inches by 15-1/2 inches.

5.1.2 Level C. Coveralls shall be preserved to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Twelve (12) coveralls of one size only, preserved as specified in 5.1, shall be packed within a fiberboard shipping container assembled, closed and reinforced conforming to class weather-resistant, grade V15c, type CF, variety DW, size 2A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat within the shipping container, one in width, two in length, and six in depth. Each coverall shall be reversed and the length of the coveralls parallel to the side panels of the container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.2 Level B. Twelve (12) coveralls of one size only, preserved as specified in 5.1, shall be packed within a fiberboard shipping container, assembled and closed, conforming to class domestic, grade 275, type CF, variety DW, size 2A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat within the shipping container, one in width, two in length, and six in depth. Each coverall shall be reversed and the length of the coveralls parallel to the side panels of the container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.3 Level C. Coveralls preserved, as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.4 Palletization. When specified (see 6.2) item packed as specified shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding methods C and D or Film bonding F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, Type IV; Type V, Class 1, Size 2; or Type VIII, fabricated from wood group I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from wood groups specified in MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

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5.5 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The items are intended for wear by carrier flight deck personnel engaged in launching aircraft by means of a catapult.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Sizes required (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, testing and approval of the first article.
e. Levels of preservation and packing required (see 5.1 and 5.2)
f. When palletization is required (see 5.4)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.5).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids (see 3.1 and 3.5).

6.4 Recycled materials. It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

6.5 Silicone compounds. Only those silicone compounds approved by the U.S. Army Natick Research, Development and Engineering Center, Natick, Massachusetts 01760, and the appropriate medical service and so listed in the invitation for bids or request for proposal, shall be considered acceptable for the related procurement (see 3.3.2.1).

6.6 Webbing, elastic. A suggested source for the polyester elastic webbing described in paragraph 3.3.7 is Elastic Corporation of America, 455 Highway 70, Columbia, AL 35051.

6.7 Reflective tape. A suggested source for the reflective tape described in 3.3.8 is 3M Company, 3M Center, St. Paul, MN 55101

6.8 Neoprene foam. A suggested source for the neoprene foam described in paragraph 3.3.9 is Rubatex Corp., P.O. Box 340, Bedford, VA 24523.

6.9 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible Military Agency.

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6.10 Subject term (key word) listing.

Clothing, aramid cloth
Clothing, carrier flight deck personnel
Uniform, aircraft launching personnel
Uniform, carrier flight deck personnel

Custodian:
Navy - NU

Review Activity:
DLA - CT

User Activity:
Navy - AS

Preparing Activity:
Navy - NU

Project No. 8415-N684

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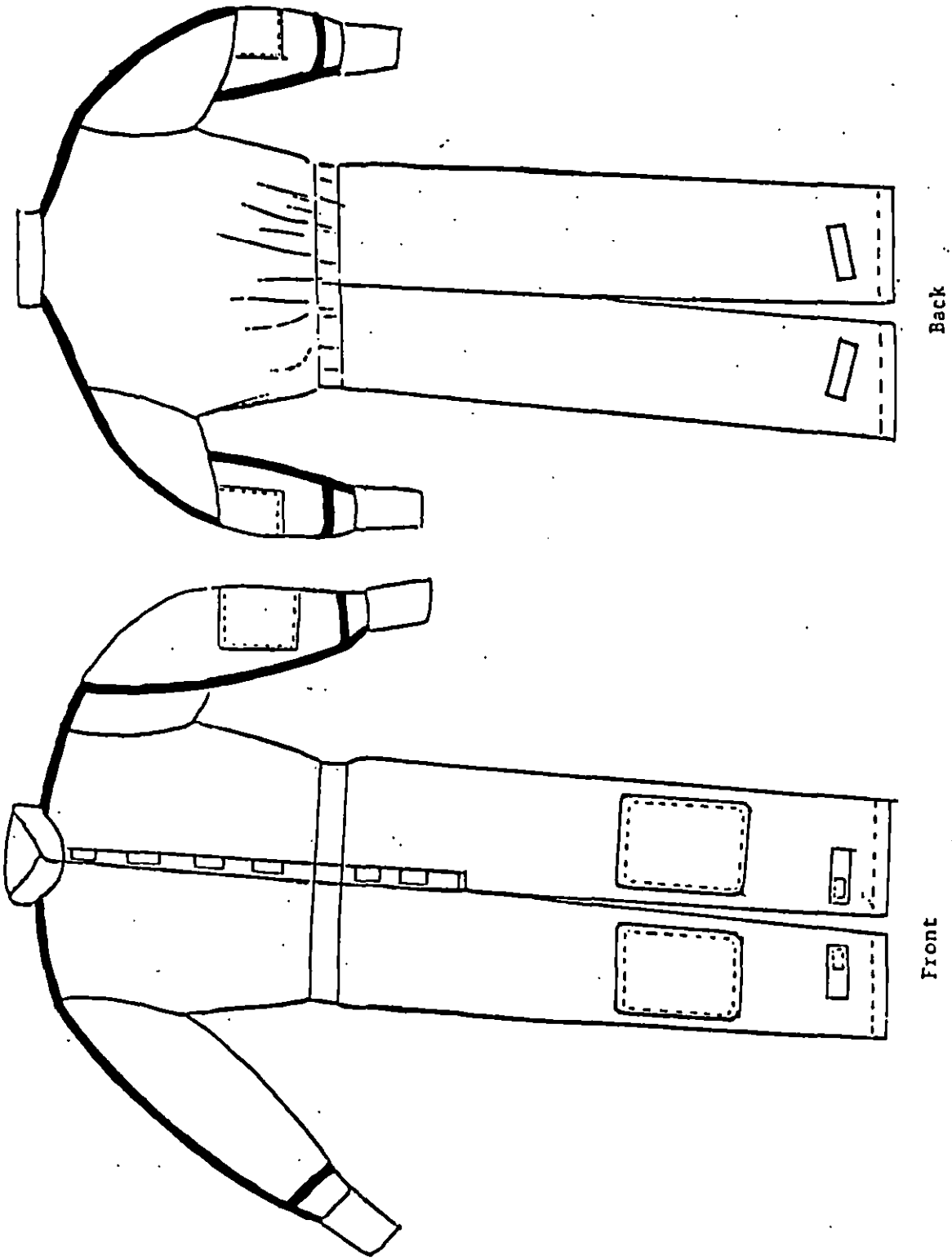


FIGURE 1 COVERALLS, CATAPULT CREWMEN'S (FLAME RESISTANT)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29627(NU)	COVERALLS, CATAPULT CREWMEN'S (FLAME RESISTANT)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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